SHIVANSH GUPTA

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About Me

Final-year B.Tech CSE student at RKGIT, Ghaziabad with strong foundations in **data structures, algorithms, and complexity analysis**, and hands-on full-stack (MERN) development experience. 350+ problems solved across LeetCode and CodeChef

Skills

Core CS: Data Structures & Algorithms, Complexity Analysis, DBMS, Operating Systems

Languages: C, Java, JavaScript

Frontend: React.js, Next.js, Vite, Tailwind CSS

Mobile: React Native

Backend: Node.js, Express.js, REST APIs

Database: MongoDB, Supabase

DevOps: Docker, AWS

Tools: Git, GitHub, Postman

Education

B.Tech – Computer Science & Engineering
RKGIT, Ghaziabad (AKTU)
2023 – 2027
CGPA: 7.3/10 (till 6th sem)

Achievements

- LeetCode Top 18% globally | 250+ problems solved
- CodeChef max rating 1850 | 300+ problems solved
- Finalist – Bharatiya Antariksh Hackathon 2026 (Team CosmicNerds)
- Contributed to 3+ open-source projects on GitHub

Experience

IT INTERN *July 2026 – Present*
NPTI (National Power Training Institute), Faridabad | Hybrid

- Conducted a QA audit of npti.gov.in, identifying and documenting 35+ bugs, broken links and content inconsistencies across the site
- Reviewed site content against GIGW 3.0 (Guidelines for Indian Government Websites) and assisted in aligning structure to compliance standards
- Identified mismatches in URLs and content between Hindi and English versions of the website

TECHNICAL INTERN *Part time*
CATI AI, Gurgaon | Remote

- Working on real-world projects involving AI, automation, REST APIs and web scraping
- Building and testing backend service integrations with structured tool-calling patterns for an AI agent system

Projects

DUALGUARD (SF2-NET)
Tech: Python, PyTorch, FastAPI, React, Node.js/Express, Tailwind CSS

- Built a dual-stream CNN (EfficientNet-B4 spatial + MobileNetV3 frequency/FFT-DCT streams) for deepfake image detection, achieving **99.72% test accuracy** and 1.0 AUC-ROC on a 140k-image dataset
- Ran cross-dataset validation (98.93% accuracy on an independent Celeb-DF v2-based set) and an ablation study isolating spatial vs. frequency vs. fused-stream contributions; results drafted into an IEEE-format paper
- Served the model via a FastAPI inference microservice with Grad-CAM heatmap explainability, integrated into a MERN-stack web app
- **Github:** github.com/Shivansh-04/dual-guard

ALGORITHM VISUALIZER
Tech: React, TypeScript, Vite, Tailwind CSS, GSAP, Anime.js

- Built an interactive visualizer for 7 sorting algorithms (Bubble, Selection, Insertion, Merge, Quick, Heap, Counting) with real-time animations at 60 FPS
- Implemented multi-language code display (JS, Java, Python) with **Big-O complexity analysis** for each algorithm
- **Github:** github.com/Shivansh-04/sort-scape-animations
- **Live:** sortingvisualizer-zeta.vercel.app

OPTIONS PRICING ENGINE
Tech: React, Vite, Node.js, Express, MongoDB, JWT, Recharts

- Built a full-stack fintech app using MERN stack for real-time options pricing and analysis
- Implemented Black-Scholes model, all 5 Greeks, and Monte Carlo simulation (10,000 paths) in pure Node.js — zero external math libraries, applying numerical/optimization methods at scale
- Features JWT auth, MongoDB Atlas, interactive Payoff Diagram and Option Chain visualizations using Recharts
- **Github:** github.com/Shivansh-04/Options-Engine
- **Live:** options-engine-red.vercel.app